Instituto Politécnico Nacional
Centro de Investigación en Computación

Mathematics for Computer Science

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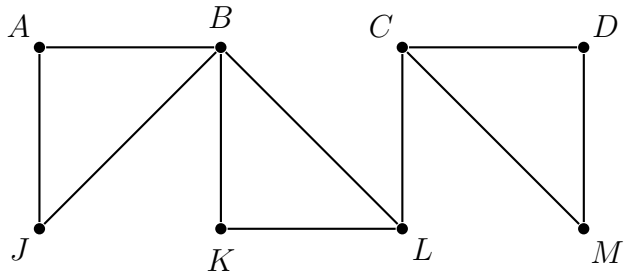
Homework 7 & 8

1. Determine whether the given graph has Euler circuit or Euler path. If so, find such circuit or path.



Solution
This graph has exactly two vertices of odd degree, namely, L and C. By a theorem we reviewed in class, since it has such vertices, then it does NOT have an Euler circuit. However, it DOES have an Euler path where its two endpoints are L and C. One such Euler path is L, K, J, A, B, L, C, M, D.

2. Determine whether the graph from the previous exercise has a Hamilton circuit or Hamilton path. If so, find such circuit or path.
Solution
The vertices A, J and K have degree 2, which means those 6 edges MUST be part of any Hamiltonian circuit (because in a Hamilton circuit, each vertex is incident with two edges in the circuit). However, node B is incident with 3 of those edges. That implies B will be visited at least 2 times, hence there is no Hamilton circuit.
On the other hand, this graph does have Hamilton paths. One such Hamilton path is A, J, B, K, L, C, M, D.